LEARNING RESOURCE GUIDE

Generative AI for Beginners | Professional Participant Workbook

Section	Section 7: Prompt Engineering Mastery
Lecture	Lecture 7.1: Introduction to Prompt Engineering

Learning Objectives

By the end of this lecture and its accompanying guide, you will be able to:

- Define prompt engineering and explain why it is essential for effective AI use.
- Describe how the structure of a prompt influences language model outputs.
- Identify the five key dimensions of a well-engineered prompt: topic, tone, scope, format, and perspective.
- Distinguish between vague and specific prompts and explain the practical impact of each.
- Apply foundational prompt engineering principles to real AI tasks in a professional context.

Introduction

Prompt engineering is the discipline of crafting inputs to AI language models in ways that reliably elicit accurate, relevant, and useful outputs. As generative AI tools become embedded in professional workflows—from drafting communications to analysing data and generating code—the ability to communicate effectively with these systems has become a meaningful workplace skill. Unlike traditional software that follows strict syntax, language models respond to natural language, making the craft of prompting both accessible and nuanced.

The core insight is that language models are pattern-completion engines. They predict what text should follow a given input based on statistical patterns learned from vast training corpora. A prompt that resembles thoughtful, professional writing tends to elicit responses of a similar calibre. A vague or ambiguous prompt may produce generic, off-target, or fabricated content. Understanding this mechanism—rather than treating the model as a magic oracle—is the foundation of effective prompt engineering.

This guide expands on the introductory framework to give you practical tools for crafting prompts in professional settings. Whether you use AI for research, writing, analysis, or creative work, the principles here apply across tools and contexts.

Core Concepts Explained

Prompt as Context-Setter

Definition: A prompt is the complete input provided to a language model. It establishes the context from which the model predicts subsequent text, including topic, tone, scope, format, and perspective.

Business relevance: Poorly formed prompts waste time through revision cycles and produce outputs requiring significant editing before they are usable. Well-formed prompts reduce iteration cost and produce first-draft quality outputs faster.

AI Practitioner relevance: For an AI practitioner—whether a product manager, analyst, or content specialist—mastering the prompt is the highest-leverage skill for day-to-day AI use. It directly determines output quality without requiring any technical background.

Real-world example: A marketing team at Unilever using generative AI for campaign copywriting found that adding explicit audience descriptions ('tone: approachable, audience: parents aged 25-40') reduced revision cycles by approximately 40% compared to bare task instructions.

Risks or limitations: Over-specifying a prompt can constrain the model's creative range when open-ended outputs are desired. Practitioners should calibrate specificity to the task: analytical tasks benefit from tight constraints; brainstorming tasks benefit from more latitude.

The Five Prompt Dimensions

Definition: Effective prompts typically address five dimensions: Topic (subject domain), Tone (formal, casual, technical), Scope (breadth and depth), Format (essay, list, code, table), and Perspective (viewpoint or role the model should adopt).

Business relevance: Each dimension is a lever. Adjusting tone alone—for example, shifting to 'explain as if to a board of directors'—can transform the utility of the output without changing the underlying information request.

AI Practitioner relevance: AI practitioners use these five dimensions as a mental checklist before submitting any complex prompt. Teams trained on this framework report faster onboarding to AI tools and more consistent output quality.

Real-world example: The BBC's digital content teams use structured prompt templates that explicitly define audience, format, and word count before submitting to generative AI systems—resulting in outputs that more reliably meet editorial standards on the first pass.

Risks or limitations: Practitioners often focus only on topic while neglecting format and tone, leading to outputs that are factually adequate but stylistically inappropriate for the intended context.

Model Behaviour as Pattern Completion

Definition: Language models generate responses by predicting the most statistically likely continuation of the prompt. They do not retrieve facts from a database or apply logical reasoning in the human sense—they extend patterns observed in training data.

Business relevance: Understanding this mechanism helps practitioners set appropriate expectations and design verification steps. Knowing the model can pattern-match a citation without that citation being real is essential for high-stakes use cases.

AI Practitioner relevance: For practitioners deploying AI in customer-facing or decision-support contexts, this concept motivates human-in-the-loop review processes and explains why prompts that establish a high-quality register tend to produce better outputs.

Real-world example: GitHub Copilot's effectiveness as a code completion tool relies on this mechanism: it predicts syntactically and semantically plausible code continuations. Teams that frame prompts as 'high-quality, well-commented code' in their context headers consistently report better suggestions.

Risks or limitations: Practitioners who treat models as fact databases rather than pattern completers are most at risk of accepting hallucinated information. This risk is highest in niche industries, recent events, and domains with sparse training data.

Prompt Iteration as a Workflow

Definition: Prompt engineering is rarely a one-shot process. Effective practitioners treat the prompt as a working hypothesis, evaluate outputs against desired criteria, and refine iteratively until quality meets the requirement.

Business relevance: Building an iteration discipline—rather than expecting perfect results immediately—is what separates practitioners who use AI effectively from those who abandon it after a poor first result. Many organisations document best-performing prompts in shared libraries.

AI Practitioner relevance: Prompt libraries and shared templates are emerging as a professional asset in AI-forward teams. Practitioners who contribute to and maintain these libraries add measurable value to their organisations.

Real-world example: Salesforce's internal prompt engineering teams maintain version-controlled prompt libraries for sales communication templates, contract summaries, and support ticket routing, updating them as model behaviour evolves with new releases.

Risks or limitations: Iteration without a clear evaluation criterion can become circular. Define what a good output looks like before iterating, so you know when to stop.

Practical Applications

The following scenarios show how prompt engineering principles apply in professional contexts.

Research Summarisation

A strategy analyst needs to summarise a 20-page industry report. A vague prompt ('summarise this document') produces a generic paragraph. An engineered prompt—'Summarise this report in three sections: key market trends, competitive threats, and recommended actions. Use bullet points. Audience: senior executives. Tone: concise and direct.'—produces a structured, immediately usable briefing. The five-dimension framework (topic, tone, scope, format, perspective) transforms the output quality.

Customer Communication Drafting

A customer success manager uses AI to draft responses to complex client complaints. Rather than pasting the complaint and typing 'reply to this', they engineer the prompt: 'You are a senior customer success manager. Draft a professional, empathetic reply that acknowledges the issue, provides a root-cause explanation, and proposes a resolution. Avoid jargon. Maximum 150 words.' The result is a near-send-ready draft rather than a starting point requiring heavy editing.

Code Documentation

A developer uses a language model to generate documentation for an internal API. A prompt specifying perspective ('You are a technical writer producing docs for developers unfamiliar with this codebase'), format ('Use Markdown with headers, parameter tables, and code examples'), and scope ('Cover function purpose, parameters, return values, and one usage example per function') produces documentation matching the team's style guide without additional formatting passes.

Best Practices

- **Define the audience explicitly.** Stating who will consume the output ('audience: non-technical executives') shapes vocabulary, assumed knowledge, and format in ways that significantly improve usability.
- **Specify format before content.** Tell the model the desired output format—list, table, essay, code block—before describing the task. Models often default to prose even when a structured format would serve better.
- **Assign a role or perspective.** Opening with 'You are a [role]' primes the model's register and domain knowledge, improving relevance without requiring additional explanation.
- **Set length or word-count expectations.** Unconstrained prompts often produce outputs that are either too long or too short. Explicit length guidance aligns output to actual need.
- **Use examples when consistency matters.** For tasks requiring specific patterns—formatting, classification, tone—include one or two examples in the prompt. This is especially effective for domain-specific tasks the model may not have seen frequently in training.

Common Mistakes and Misconceptions

Misconception: Assuming the model understands implicit context.

Correct approach: Make all relevant context explicit. The model has no access to your prior knowledge, organisational context, or unstated assumptions. What seems obvious to you is invisible to the model unless you state it.

Misconception: Accepting the first output without evaluating it against criteria.

Correct approach: Define what a good output looks like before submitting the prompt. Evaluate the first output against those criteria. If it falls short, identify which dimension is misaligned and refine.

Misconception: Using identical prompts across different models.

Correct approach: Different models (GPT-4, Claude, Gemini) have different strengths and instruction-following behaviours. A prompt optimised for one model may underperform on another. Test prompts on the specific model you are deploying.

Misconception: Over-relying on AI outputs without verification.

Correct approach: Even well-engineered prompts can produce hallucinated facts, especially for niche or recent topics. Build a verification step into any workflow where accuracy is consequential.

Prompt Engineering Checklist: Five-Dimension Prompt Review

Before submitting any professional prompt, run through this checklist to ensure each dimension is addressed. Skipping dimensions is the most common source of inadequate outputs.

Dimension	Question to Ask	Example Value
Topic	Is the subject/domain clearly stated?	Summarise Q3 sales performance
Tone	Is the desired register specified?	Formal, executive-level
Scope	Is the depth and breadth defined?	Top 3 trends only; 200 words max
Format	Is the output structure specified?	Bullet points under 3 headers
Perspective	Is a role or viewpoint assigned?	You are a senior financial analyst

How to use this: *Complete all five rows before submitting high-stakes prompts. For routine tasks, at minimum define Topic, Format, and Scope.*

Knowledge Check

Test your understanding before moving on. Answers are shown below each question.

1. A marketing manager asks an AI tool to 'write something about our new product launch.' The output is generic and unusable. What is the most likely cause?

A. The AI model is not capable of marketing tasks.

B. The prompt lacks specificity across key dimensions such as audience, tone, and format.

C. The product launch information was not in the model's training data.

D. Marketing prompts require code to execute properly.

Answer: B. The prompt lacks specificity across key dimensions such as audience, tone, and format.

Why: *Vague prompts produce generic outputs because the model has insufficient context to predict a useful continuation; the five-dimension framework directly addresses this.*

2. An analyst wants the AI to produce a concise executive summary rather than a long essay. Which prompt addition is most likely to achieve this?

A. 'Be brief.'

B. 'Summarise only the important parts.'

C. 'Produce a 150-word executive summary using three bullet points: findings, risks, and recommendations.'

D. 'Write a short version.'

Answer: C. 'Produce a 150-word executive summary using three bullet points: findings, risks, and recommendations.'

Why: *Explicit format and length constraints reliably shape output more than vague qualifiers like 'brief' or 'short'.*

3. A language model responds to a legal question with a highly confident but inaccurate citation. What explains this behaviour?

A. The model deliberately fabricated the citation to mislead.

B. The model pattern-matched a plausible citation format from training data without verifying the source.

C. The prompt was too long, causing the model to confuse itself.

D. Legal content is blocked by the model's safety filters.

Answer: B. The model pattern-matched a plausible citation format from training data without verifying the source.

Why: *Models generate statistically plausible continuations and do not verify facts, so citation-shaped text can be produced even when no real citation exists.*

4. Which approach best represents the prompt iteration discipline recommended for professional AI use?

A. Submit the prompt once; if the output is poor, switch to a different AI tool.

B. Define success criteria, evaluate the output, identify which dimension is misaligned, and refine.

C. Submit the same prompt multiple times until a good output appears by chance.

D. Use the model's suggested prompt revisions without evaluation.

Answer: B. Define success criteria, evaluate the output, identify which dimension is misaligned, and refine.

Why: *Systematic iteration against defined criteria is the professional discipline that separates effective AI practitioners from casual users.*

Reflection Questions

1. Think of a task you regularly perform at work. How would you apply the five-dimension framework to engineer a prompt for it? Write out each dimension explicitly.
2. Where in your current workflows could poorly engineered prompts create professional risk? What verification step would you add?
3. How does the concept of 'prompting as pattern-completion' change how you approach an AI tool compared to a search engine?
4. What shared prompt library would be most valuable to your team? What categories of prompts would it contain?
5. If you were onboarding a colleague to AI tools for the first time, which of the five prompt dimensions would you teach first, and why?

Key Takeaways

- Prompt engineering is the practice of crafting AI inputs to reliably produce high-quality, relevant outputs—accessible to non-technical practitioners.
- Language models are pattern-completion engines, not fact databases. Prompts resembling professional writing tend to elicit similar outputs.
- The five prompt dimensions—Topic, Tone, Scope, Format, and Perspective—provide a practical framework for structuring any professional prompt.
- Vague prompts produce generic outputs; specific prompts with explicit context, format, and length constraints produce immediately usable results.
- Prompt engineering is an iterative discipline: define success criteria, evaluate outputs, and refine systematically rather than hoping for a lucky result.
- Shared prompt libraries are an emerging professional asset; teams that document and maintain effective prompts compound their AI productivity over time.
- Verification remains essential—even well-engineered prompts can produce hallucinated content in niche or high-stakes domains.

Recommended Next Actions

6. Select one recurring professional task and apply the five-dimension checklist to engineer a prompt for it. Compare the output to your previous approach.
7. Create a simple personal prompt library for the three prompts you use most frequently. Refine them over the next two weeks.
8. Practice identifying which dimension is causing a poor output whenever you receive one. Retrain your instinct to diagnose before resubmitting.
9. Review any AI outputs you sent externally in the past month. Identify one that would have benefited from better prompt engineering and redesign it.